

Mandatory Assignment 2

April 3, 2026

General remarks

- You have to hand in one (!) .qmd (Quarto) script with commented source code in Python or R and one (!) .pdf report in which you describe your methods and results (e.g., with figures, tables, equations, etc.). The fitted model objects from exercise 2d are also required for peer review.
- The report (font size 12pt, use the provided template) may not exceed 7 pages including figures, tables, and other appendices
- The source code has to create every figure, table, and number used for your report
- When reporting numbers in your report, please consider a meaningful rounding scheme (there is usually no need to report six digits after the comma)
- Within the report, describe every significant step and make sure that tables and figures are self-explanatory by providing meaningful captions and variable names. The reader will not consult your code — descriptions, captions, and variable names must be self-contained.
- Meaningful code comments explaining the purpose of each step improve both the quality of peer feedback and your final grade
- **Minimum requirement:** the code must run without error or interruption. Ensure the code loads all required packages at the beginning of the script (you can assume that the user has the packages available locally, so do not include `install.packages()` commands). Attempt to solve every task. Document your struggles if you do not finish an exercise. **Hand-ins that do not fulfill this minimum requirement will not receive peer feedback and will not be considered for the exam qualification requirements**
- The assignments can be written individually or by groups of a maximum of three students
- Observe the university's plagiarism rules; for group submissions, note the additional rules for co-written work
- All parts must be answered in English, including comments in the code

The deadline to hand in the files on Peergrade via Absalon is **Sunday, April 12th, 8 pm.**

Empirical Asset Pricing via Machine Learning (inspired by Gu et al. [2020])

Predicting stock returns matters for understanding risk premia. Gu et al. [2020] show that machine learning forecasts generate significant economic gains for investors. Their approach helps measure risk premia more reliably and sheds light on the economic mechanisms behind asset pricing.

In this assignment, you replicate a part of their research and analyze the results. You receive a portion of the original sample with 3045809 individual stock-month observations. The overarching research questions your report addresses are: Can machine learning methods identify stock characteristics that predict cross-sectional return variation? Do these predictions translate into economically meaningful investment strategies after accounting for real-world frictions?

Exercise

The four exercises build on each other. Your variable selection in Exercise 1 determines the inputs for the models in Exercise 2, which produce the forecasts for the portfolios in Exercise 3. Exercise 4 then reveals which variables actually drive the predictions. Choose carefully in Exercise 1, as changing your selection later requires re-running the entire pipeline.



Tip

Practical note: Training machine learning models on large panels can be computationally intensive. If training is slow, consider reducing the hyperparameter grid or the number of characteristics. Consult the discussion board on Absalon if you run into technical difficulties. To render a Quarto document that includes heavy computations, you can set `cache: true` in the YAML header. Code chunks will then only be re-evaluated if their content changes, saving time when editing the report text.

1. *Goal: build a clean, well-understood dataset for return prediction.* Read in the file `stock_characteristics_monthly.parquet` from Absalon and prepare the dataset. The file contains 94 stock characteristics from Gu et al. [2020] (GKX, henceforth) and 13 macroeconomic predictors from the Goyal and Welch [2008] data library. The file also contains 73 industry classifications (`sic2`), excess returns (`ret_excess`), lagged market capitalization (`mktcap_lag`), and firm and date identifiers (`permno`, `date`). Stock characteristic columns are named `characteristic_*` and macroeconomic predictor columns `macro_*`.

GKX model an asset's excess return as

$$r_{i,t+1} = E_t(r_{i,t+1}) + \varepsilon_{i,t+1} \text{ where } E_t(r_{i,t+1}) = g(z_{i,t})$$

for stock $i = 1, \dots, N_t$ in month $t = 1, \dots, T$. The predictor vector $z_{i,t}$ collects stock characteristics and macroeconomic variables. The goal is to find a function $g(\cdot)$ that maximizes out-of-sample explanatory power for realized excess returns $r_{i,t+1}$.

Conduct the following preparatory steps:

- a. Remove all observations before January 1st, 2000. Select up to ten influential stock-level characteristics and up to four macroeconomic variables. Justify your selection. You can refer to Figure 5 and Table 4 of GKX.

! Important

Timing convention: Each row already aligns predictors $z_{i,t}$ with the next month's excess return $r_{i,t+1}$. You do **not** have to lead or lag any variable.

- b. Summarize the included variables with meaningful statistics or illustrations. You can also use the industry classification in column `sic2`. Handle missing values in your selected predictors and document your approach (e.g., drop incomplete rows, impute with cross-sectional median).
 - c. Use `tidymodels` or `sklearn` to prepare the feature matrix. Include all interactions between stock-level characteristics and macroeconomic state variables. Concretely, the feature matrix consists of [stock characteristics, stock characteristics $\times$ macro variables]: with 10 characteristics and 4 macro variables, this gives $10 + 40 = 50$ continuous features (plus SIC dummies). In R, use `step_interact(terms = ~ contains("characteristic"):contains("macro"))` (consistent with the lecture slides); in Python, construct interactions with a manual loop or `PolynomialFeatures(interaction_only=True)`. Apply one-hot encoding to the industry classification `sic2`. If many SIC2 codes appear, consider recoding them into the 10 major SIC division categories to keep dimensionality manageable.
2. *Goal: gain hands-on experience with model selection, tuning, and the bias-variance trade-off.* Choose two machine learning methods of your choice (penalized linear models, tree-based models, neural networks) to forecast excess returns.
- a. For each of your two methods separately, very briefly describe the objective function, the model structure, and the hyperparameters that require tuning.
 - b. Split the dataset into training, validation, and test samples. The split **must respect time**: all training observations must precede all validation observations, which must precede all test observations — a random split would leak future information into training. Use the same split for both methods. Reserve the most recent 20% of observations for out-of-sample testing.¹ Report the date ranges for each set. Justify your choices for the relative sizes of training and validation sets.²
 - c. Tune the hyperparameters, fit each model, and select the best specification based on validation performance. Show how model complexity affects predictive accuracy in training and validation sets. State the final hyperparameter values and your selection criterion.
 - d. Store each final model object to disk. In R, use the `butcher` package to reduce model size before saving with `write_rds()`. In Python, use `joblib`. Name the files `model_<method>.rds` or `model_<method>.joblib` (e.g., `model_elastic_net.rds`). Include in your report the code your peer reviewer needs to load each model.
3. *Goal: translate statistical forecasts into an investment strategy and assess whether predictability survives real-world frictions.*
- a. Generate predicted excess returns on the validation and the test set for each model.

¹You may **not** use test observations for model fitting or hyperparameter selection.

²Contrary to GKX, you do not need to roll the estimation window over time. Defining the splits once suffices.

- b. For each month, sort stocks into decile portfolios by predicted excess returns. Use `mktcap_lag` for value-weighted returns within each decile.
 - c. Form a long-short portfolio (long top decile, short bottom decile) to create a zero-net-investment strategy.
 - d. Report **annualized** mean excess return, Sharpe ratio, and CAPM alpha with t-statistic for both the validation and the test period.³ As a benchmark, report the market portfolio’s annualized return and Sharpe ratio over the same periods. Discuss any difference in performance between the two periods: what does it suggest about the stability of the return predictability?
 - e. Propose and report a suitable measure of monthly portfolio turnover. Discuss what the turnover level implies for the net-of-cost profitability of the strategy.
 - f. Compare the two machine learning models: which generates higher alpha, which has lower turnover, and which would you prefer as an investor?
4. *Goal: move from prediction to interpretation and connect data-driven results back to financial theory.*
- a. Measure variable importance following GKK: for each predictor, set it to its mean (i.e., zero after standardization) in the test set, regenerate predictions, rebuild the long-short portfolio, and record the drop in annualized Sharpe ratio relative to the baseline from Exercise 3d. A large drop indicates that the predictor contributes substantially to portfolio performance. Present the top 3 most important variables per model.
 - b. Compare importance rankings across models. Do they agree? Where do they disagree, and why might this reflect the models’ different functional forms?
 - c. **Connect findings to financial theory.** For the most important variables, explain the economic mechanism: *why* should this characteristic predict future returns? Does the pattern align with known risk premia (value, momentum, size, profitability) or suggest behavioral explanations? Reference GKK Figure 5 or Table 4 where relevant.

References

- Amit Goyal and Ivo Welch. A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508, 2008. doi: 10.1093/rfs/hhm014.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.

³If you need external data (e.g., factor returns), download it programmatically in your code. Do not share any additional data files with your peer reviewer; only the fitted model objects should be uploaded.